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Inferring security-relevant properties from Java documentation

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Abstract

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Chapter 1

Introduction

Static application security testing (SAST) applies static program analysis to the security of software projects. [MORE INFO ABOUT STATIC ANALYSIS HERE](#). Static analysis is performed on the source code, compiled byte-code, or some other representation of the source code [1]. Popular SAST tools include: SonarQube [2], SpotBugs [3], Snyk Code [4], and Facebook’s Infer [5]. SAST tools have become widely used at all stages of the software development life-cycle. Furthermore, they have been demonstrated to be highly effective in identifying defects in code compared to other methods, such as manual inspection [6].

Unfortunately, SAST tools are not perfect. From Rice’s Theorem, it is known that any non-trivial property about programs is reducible to the halting problem [7]. That is, a ”perfect” analysis is not always possible and approximations must be made in order for the analysis to terminate, resulting in an imperfect output. False positives (where safe code is incorrectly labelled as a defect) and false negatives (where the analysis misses a genuine fault) [8] are common in all static analysis tools. False positives are not ideal as manual inspection is required to separate genuine defects from false positives, leading to an increased time and labour cost, and false negatives mean genuine defects remain undetected. To reduce the rate of false negatives, manual configuration of the SAST tools for specific software projects is often required [9].

For example, to perform an effective taint analysis, a custom configuration of sources, sinks and sanitisers for a particular project is often required. Creating this manual configuration is a time-consuming and resource-intensive process. As such, methodologies to automatically label methods as sources, sinks, and sanitisers may enable existing taint analysis tools to operate more effectively. It should also be noted that in cases where a closed-source library is being used in a software project, it may be impossible to determine how the API methods should be labelled for the taint analysis.

Deep learning, a subtopic within the broader machine learning, has recently increased in popularity. What was once a niche topic reserved for academics with access to computational resources far out

of reach by most has become arguably one of the most influential technologies of the modern age. Deep learning has fundamentally changed our everyday life, with applications ranging from targeted advertising to identifying COVID-19 cases from lung CT scans [10]. Natural-language processing (NLP) is one often overlooked application of deep learning, but it has seen tremendous growth in recent years.

NLP has existed in one form or another for decades. In the early days of NLP, work was painstakingly done using handwritten rules before moving to probabilistic methods in the 1990s and then to deep learning in the 2010s [11]. With tech conglomerates such as Google and Microsoft publicly releasing large pre-trained models, NLP has never been more accessible. While machine translation and speech recognition are among the most popular applications of NLP, one application that has not been explored thoroughly in the literature is software security.

While being an informal description of code, Javadoc (i.e., natural language descriptions of code) can contain information relevant to software security. Using powerful pre-trained models such as CodeBERT that understand the semantic connection between code and natural language [12], it may be possible to extract security-relevant properties from Javadocs and use this information to configure existing SAST tools.

1.1 Topic Definition

Static application security testing (SAST) tools often require additional configuration to conduct a practical security analysis. Typically, this configuration is done manually and demands a substantial time, labour, and skill cost to be done effectively. While being an informal natural-language description of a Java method, Javadoc may contain valuable information for the configuration of SAST tools. With powerful pre-trained natural language processing models being made publicly available, it may be possible to train a deep learning model to extract relevant security properties from Javadocs so that information can be used to configure existing SAST tools.

One such pre-trained model of particular relevance is Microsoft's CodeBERT. CodeBERT has been trained on natural language and numerous programming languages (including Java). CodeBERT understands the semantic connection between natural language and programming languages, making it suitable for downstream tasks involving the relationship between natural language and code. [12].

It is hypothesised that CodeBERT can be fine-tuned with additional data to extract relevant security properties from Javadoc. This information can be used to configure existing static analysis tools. For example, taint analysis requires a list of methods labelled as sources, sinks, and sanitisers. Taint analysis tools often have a default configuration with popular and frequently used APIs. However,

security-critical methods are not included in the default configuration, particularly in large projects, which in turn leads to an incomplete analysis being conducted. Manual inspection is required to label additional methods and add them to the taint analysis configuration. Naturally, this is a labour-intensive process with high-skill requirements. Furthermore, in cases where a closed-source library is being used — where only the binary and documentation are available — it may be difficult or impossible to label these methods correctly. With sufficient training, it is believed codeBERT can adequately label security sensitive methods.

Chapter 2

Deep Learning

To provide context for this project, a high-level overview of deep learning will be provided.

2.1 Deep Learning Model Architectures

There are a large variety of deep learning models, each designed for different purposes and with different strengths and weaknesses. Given how computationally expensive training and fine-tuning models is, ensuring that an appropriate architecture is used (indirectly chosen when selecting a pre-trained model) is an important decision key for this project's success. In this section, we will discuss and compare common deep learning model architectures at a high-level in order to determine an appropriate selection for this project.

2.2 Simple Neural Network

Here, we will outline a basic neural network model to familiarise ourselves with the terminology that will be used later in the report, as well as the project moving forward.

Figure 2.1: A Simple Neural Network

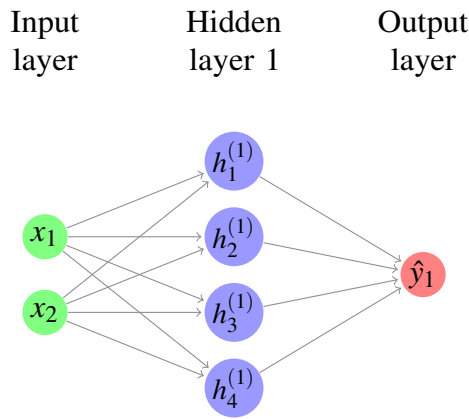


Figure 2.2: Diagram of a singular neuron

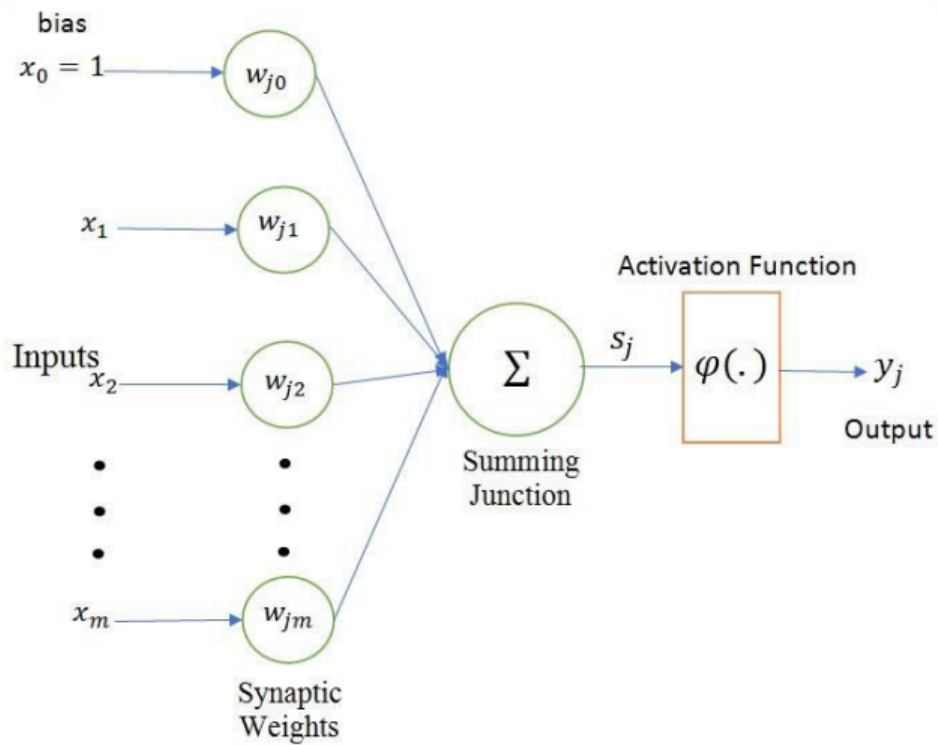


Figure 2.1 shows a simple neural network. A neural network consists of an input layer, which is a vector. Transforming the raw data (e.g. images, natural language, Java documentation, etc.) into a vector is known as *encoding*. How encoding is performed differs for all models, and is a consequential decision when designing a model. Common methods for encoding include: integer encoding, one-hot encoding, and word embeddings. Word embeddings are of particular importance as they are used by CodeBERT, the pre-trained model that will be used for this project.

Each neuron sums all the values of the neurons connected to it (generally, this is *all* the previous neurons.). The edges between neurons have *weights*, which are multiplied against the value before being added to the neuron. Furthermore, each neuron has a *bias* which is added to the neuron after each calculation. Figure 2.2 shows this graphically.

When this sum is generated, it is then passed into an *activation function*. The activation function stabilises the value of each neuron, ensuring they do not drop to zero or grow exponentially. These two issues are commonly referred to as the *vanishing gradient* and *exploding gradient* problems, respectively. The choice of activation function is important. There is a trade-off between the computational cost of activation functions, and how effective they are. Common activation functions include: the sigmoid function, hyperbolic tangent function, and gaussian function.

2.2.1 Recurrent Neural Networks

Processing language is difficult. The meaning of words is not just determined by itself, but also in what context the word is being used in. That is, the meaning of a word is not just determined by itself, but also by the words that surround it. The simple neural network introduced above would not be able to handle such a situation as it has no “memory” of previous inputs, i.e. it lacks contextual information.

Recurrent Neural Networks are one methodology for neural networks to have “memory” and thereby consider what context an input is being used in. Essentially, the state of an RNN at any input is determined by all previous inputs. Having “memory” of previous inputs allows RNNs to perform complex tasks on sequence data such as: voice recognition computer vision or natural language tasks such as translation.

While RNNs have historically proved useful for NLP tasks [13], they have numerous downsides. Perhaps the key issue with RNNs is that they are notoriously difficult to train, due to the *vanishing gradient* and *exploding gradient* problems. Difficulty here meaning both that they are computationally expensive to train, and can often produce inaccurate results for long sequences of inputs. Most solutions to these issues slow training considerably [14]. LSTM (Long short-term memory) networks are a class of RNN conceived to address the vanishing gradient and exploding gradient problems present in simple RNNs [15].

2.2.2 Long short-term Memory Networks

Long short-term memory networks (LTSMs) are a subclass of RNN designed to address the vanishing gradient and exploding gradient issues present in simple RNNs. Describing the precise methodology in which this is done is outside the scope of this report. However, put simply, this is achieved with “gating units” which are a more sophisticated method for calculating the hidden-state of a cell (Where the hidden-state of a cell is the “memory” the neural network uses to understand the context of an input [16]).

These gating units, in addition to using the activation function that simple RNNs use, have to calculate a variety of additional values. These extra computations accumulate and make LTSMs far

more computationally intensive to train than their generic RNN counterparts. However, this is not without benefit. LSTMs are significantly more accurate for longer sequences compared to generic RNNs [17].

2.2.3 Transformers

The goal of the transformer model, introduced by Google Brain in 2017, was to handle sequential data, just as RNNs do, but without the training-time limitations . Ultimately, this goal was achieved as transformer models were demonstrated to perform significantly better than RNNs while simultaneously requiring less training time [18].

Long training times for RNNs are, ultimately, caused by the fact that input must be processed sequentially. Transformers, however, do not have this limitation. They can process multiple inputs simultaneously and those inputs do not have to be in order. It is because transformers are parallelisable that they have quicker training times [18].

Chapter 3

Conclusion

Conclude your thesis.

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Appendix A

Appendix

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